

Lecture 06

Regression Analysis & Introduction to Machine Learning Techniques

Review on Lecture 5

- ❖ Probabilistic Modeling in Hydrology
- ❖ Flood Frequency Analysis
 - ◆ Empirical Method (No Distributional Assumptions)
 - ◆ Analytical Method (Distributional Assumptions)
- ❖ More Distributions for Extreme Events
 - ◆ Generalized Extreme Value (GEV) Distributions
 - ◆ Peaks-over-threshold (POT) Approach (=Partial Duration)
 - Generalized Pareto Distribution (GPD)
 - ◆ How to Choose the Threshold u ? Mean Residual Life Plot

Today's Topic: Lecture 06

Regression! (회귀분석)

❖ Part I

- ◆ Simple Linear Regression
- ◆ Multiple Linear Regression
- ◆ How to estimate the parameters? OLS/GD

❖ Part II

- ◆ Logistic Regression: Sigmoid Function (Relationship to ANNs?)
- ◆ Introduction to Few Machine Learning Techniques

Basic Definitions for Lecture 06

❖ Regression (회귀분석)

◆ 기본 개념 (수식) $\hat{y} = f(x) \approx y$

◆ Objective:

- 확보된 관측 자료를 통해 모델을 만들고 (매개변수를 추정하고),
- 이후 추가적인 관측을 통해 생기는 결과를 예측하고자 함

❖ Explanatory Variable (독립변수, 설명변수; denoted as x)

- ◆ a measure of something that is driving force to the system
ex) time, rainfall, streamflow, geographic coordinate, ...

❖ Response (or dependent) Variable (종속변수; denoted as y)

- ◆ variable to be estimated using the explanatory variable(s)
ex) streamflow, solute concentration, ...

Simple Linear Regression Model

$$y_i = \beta_0 + \beta_1 x_i + \epsilon_i \quad i = 1, 2, \dots, n$$

- ◆ y_i = i th observation of the response (or dependent) variable
- ◆ x_i = i th observation of the explanatory variable
- ◆ β_0 = intercept (***parameter to be estimated***)
- ◆ β_1 = slope (***parameter to be estimated***)
- ◆ ϵ_i = random error or residual for the i th observation (오차, 잔차)
 - Random variable which is independent of x_i
 - $\epsilon \sim N(0, \sigma^2)$: 별도의 분산 σ^2 을 가지고 정규분포를 따르는 RV
- ◆ n = sample size

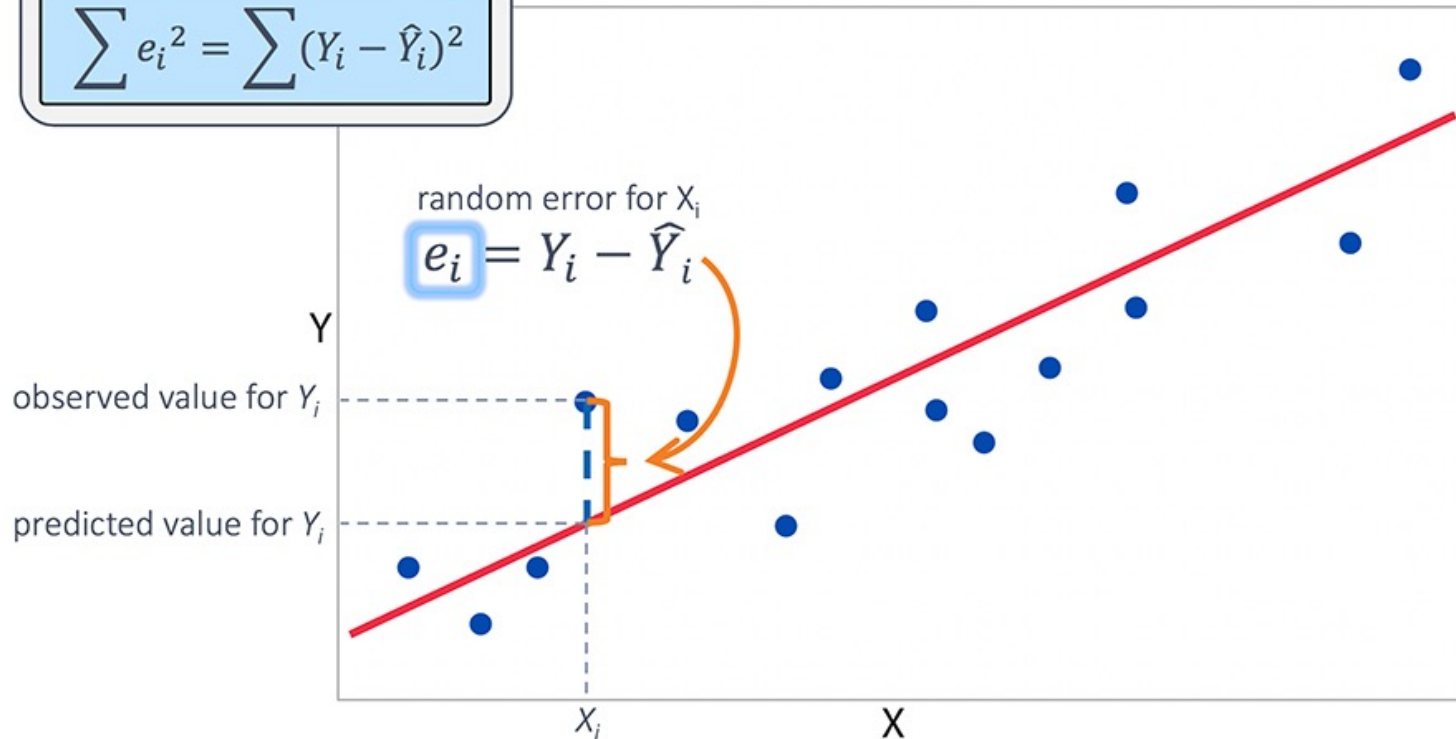
❖ Preview: AR(1) Model in Time Series

$$y_t = \mu + \phi_1(y_{t-1} - \mu) + \epsilon_t$$

Simple Linear Regression Model

Method of Least Squares

$$\sum e_i^2 = \sum (Y_i - \hat{Y}_i)^2$$



Consider x as t (Trend Detection)

❖ Are conditions changing?

◆ Here, conditions refer to mean, variance, ...

◆ Possible types of change: linear, monotonic, stepwise, expon

❖ Is there a linear trend in the mean over time?

$$y_t = \beta_0 + \beta_1 t + \epsilon_t \quad \epsilon_t \sim N(0, \sigma^2)$$

Consider x as t (Trend Detection)

$$y_t = \beta_0 + \beta_1 t + \epsilon_t \quad \epsilon_t \sim N(0, \sigma^2)$$

❖ Assumptions

- ◆ t (time) is deterministic
- ◆ parameters β_0 and β_1 are fixed & unknown
- ◆ ϵ_t is stochastic (RV)
- ◆ y_t is stochastic

Consider x as t (Trend Detection)

❖ Estimation of Parameters (via MLE)

$$L(\beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n f(y_t | \beta_0, \beta_1, \sigma^2, t)$$

By setting

$$\frac{\partial l}{\partial \beta_0} = 0 \quad \frac{\partial l}{\partial \beta_1} = 0 \quad \frac{\partial l}{\partial \sigma^2} = 0$$

We get

$$\widehat{\beta_0}$$

$$\widehat{\beta_1}$$

$$\widehat{\sigma^2}$$

Consider x as t (Trend Detection)

- ❖ By substituting $y_t = \beta_0 + \beta_1 t + \epsilon_t$ into $\widehat{\beta_1}$ expression,
We get

Multiple Linear Regression Model

$$y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \cdots + \beta_k x_k + \epsilon \quad i = 1, 2, \dots, n$$

❖ This case,

- ◆ k explanatory variables: $x_1, x_2, \dots, x_k$
- ◆ assumed to be n observations of the k explanatory variables and of the dependent variable y (subscript i omitted)

❖ Examples

- ◆ Streamflow basin characteristics model
 - Mean flow, 10-year flood (y) ~ drainage basin area, average basin altitude, percentage of basin forested (x)
- ◆ Basin yield-basin characteristic model
 - Basin dissolved solids yield (y) ~ average rainfall, percentage of the basin area underlain by carbonate rocks, basin population (x)

Determining the Model Parameters

❖ Estimation of parameters of regression models (β)

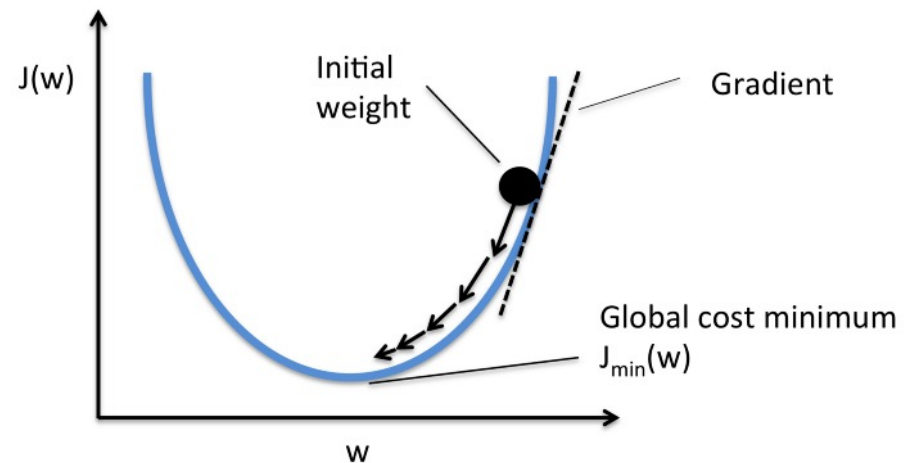
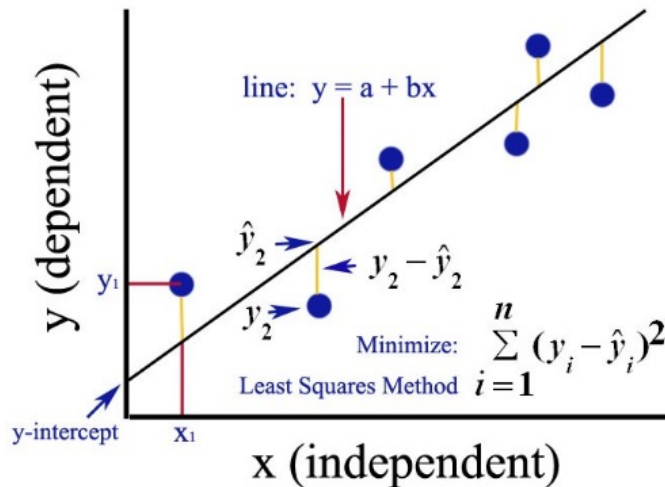
◆ Ordinary Least Square (OLS): Minimize

- Non-iterative

$$\sum_{i=1}^n (y_i - \hat{y}_i)^2$$

◆ Gradient Descent (Optimizer, GD)

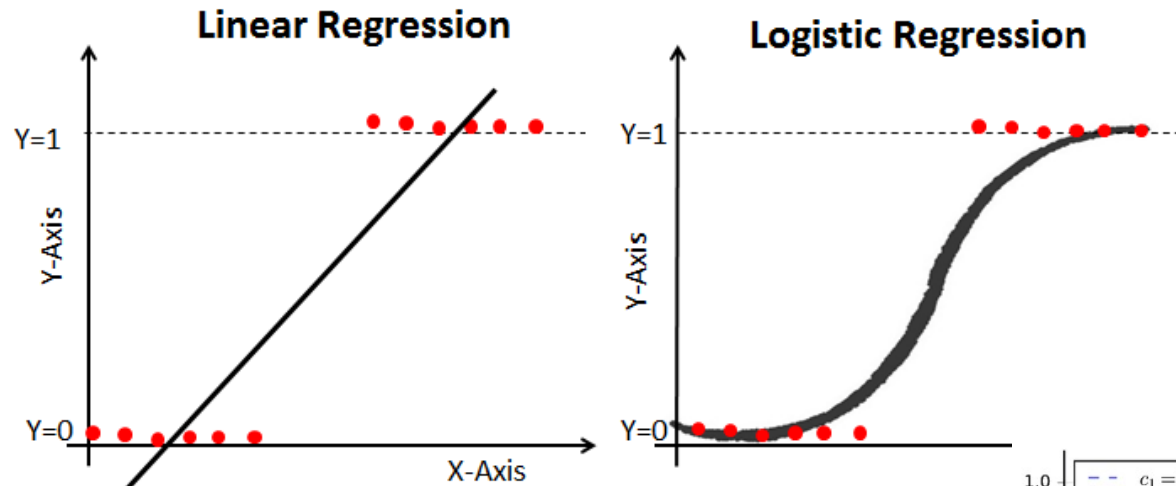
- Iterative, Learning rate α should be determined



Part II: Logistic Regression

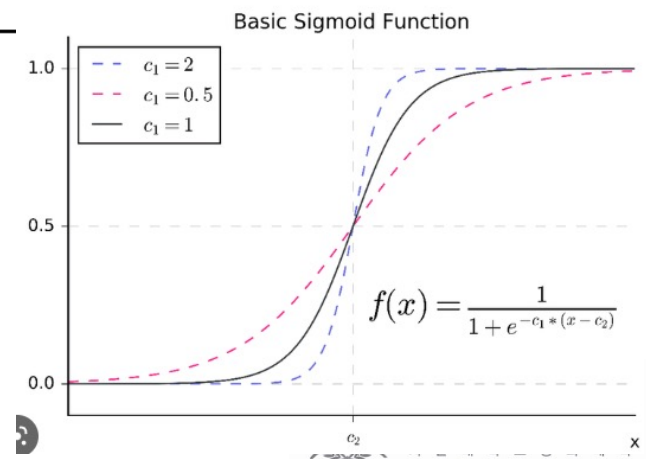
❖ Definition:

Statistical analysis method to predict a binary outcome, based on prior observations of a data set



❖ Logistic Function (Sigmoid Function)

$$F(X) = \frac{1}{1 + e^{-x}} = \frac{e^x}{e^x + 1}$$



Few Machine Learning Techniques

❖ Introducing today:

◆ Artificial Neural Network (ANN)

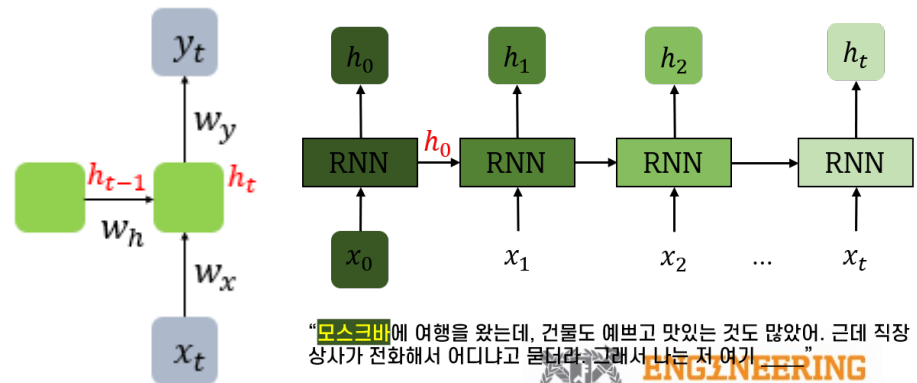
- ⊕ known to mimic highly non-linear & complex systems well
- ⊖ any information about the sequential order of the inputs are lost

◆ Deep Neural Network (DNN)

- Additional number of hidden layers between input & output layers

◆ Recurrent Neural Network (RNN)

- ⊕
- ⊖ Long-term dependency
(Only 10 or less time steps)



“모스크바에 여행을 왔는데, 건물도 예쁘고 맛있는 것도 많았어. 근데 직장 상사가 전화해서 어디냐고 묻더라. 그래서 나는 저 여기.”

Few Machine Learning Techniques

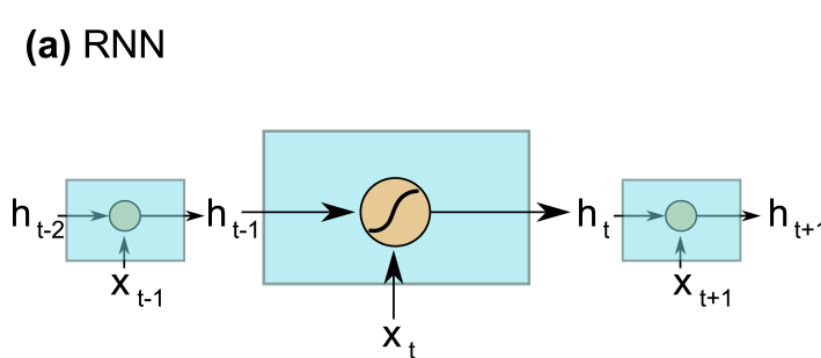
❖ Long-Short Term Memory (LSTM)

⊕ overcomes the problem of the traditional RNN of learning long-term dependencies representing hydrological catchments

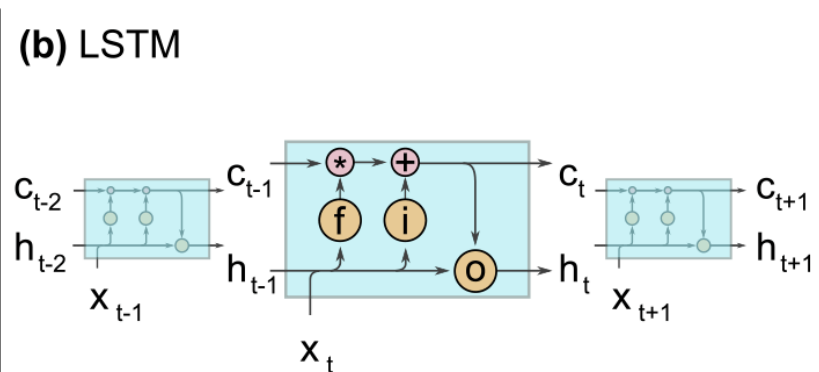
❖ LSTM compared to RNNs

- ◆ Additional **cell state (or cell memory)** where information are stored
- ◆ Forget/Input/Output gates (ⓕ ⓘ ⓘ) that control the information flow within the LSTM cell

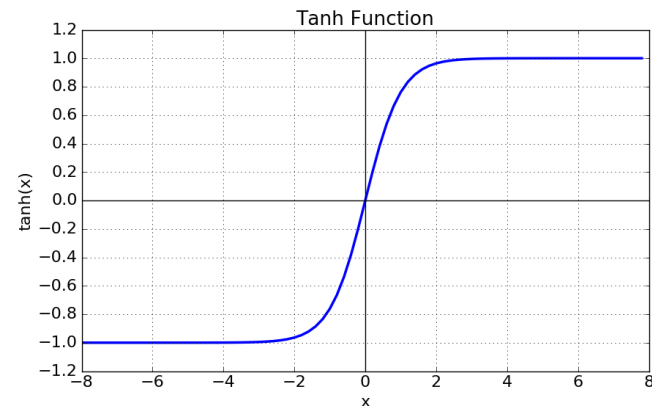
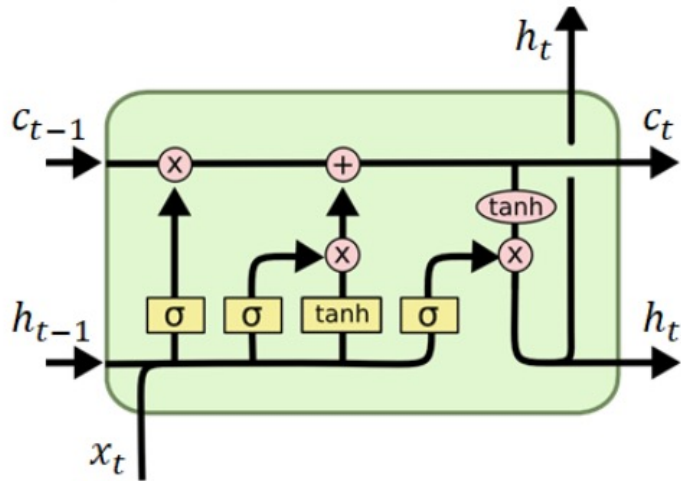
(a) RNN



(b) LSTM



Few Machine Learning Techniques



- ❖ Forget Gate: 과거의 정보를 얼마나 가져올지 결정
- ❖ Input Gate: 현재의 정보를 얼마나 저장할지 결정
- ❖ Update Gate: 과거의 Cell State (c_t)를 새로운 State로 업데이트
- ❖ Output Gate: 현 시점에서의 Cell State를 얼마나 내보낼지 결정

Examples of LSTM in Hydrology

[Article](#)[Peer review](#)[Metrics](#)[Related articles](#)

Research article

22 Nov 2018

Rainfall–runoff modelling using Long Short-Term Memory (LSTM) networks

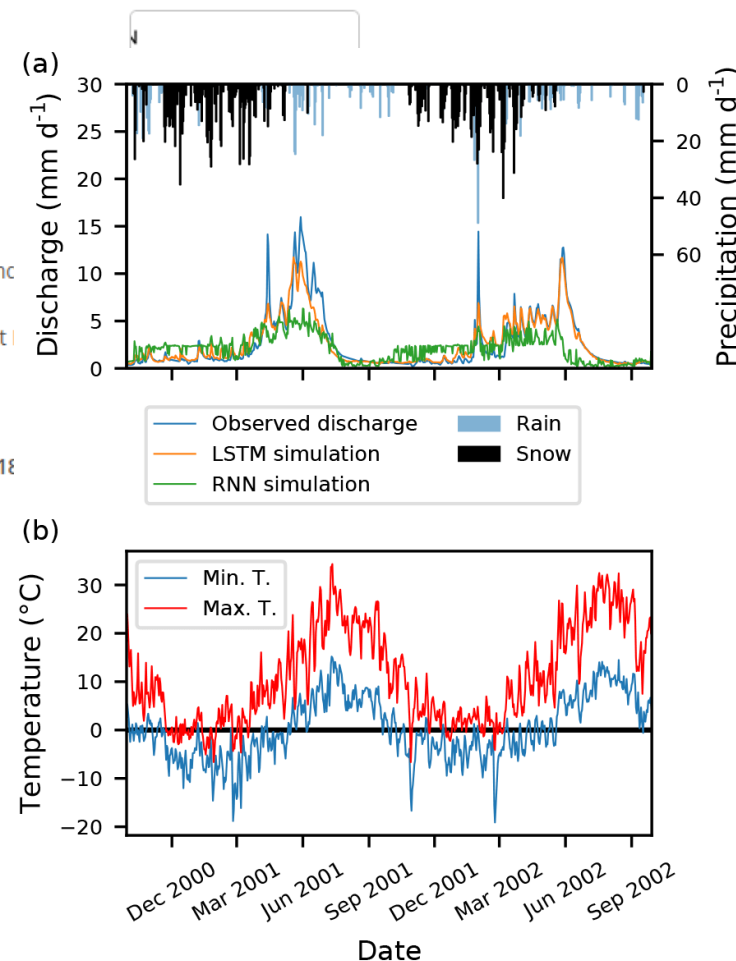
Frederik Kratzert^{1,*}, Daniel Klotz¹, Claire Brenner¹, Karsten Schulz¹,
and Mathew Herrnegger¹

¹Institute of Water Management, Hydrology and Hydraulic Engineering, University of Natural Resources and Sciences, Vienna, 1190, Austria

* Invited contribution by Frederik Kratzert, recipient of the EGU Hydrological Sciences Outstanding Student

Correspondence: Frederik Kratzert (f.kratzert@gmail.com)

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Future Schedule Preview

❖ 다음 주 강의 (2022-11-22 화요일 18:30-21:20)

◆ *Final Exam Preparation (Optional)*

◆ Multiple Choice, Short Answers, Long Answers/Problem Solving

◆ Keywords for the Test

- Conditional probability, Bayesian theorem, Extreme value analysis, MLE, Linear regression, ...

❖ Future Schedule (~End of Semester)

◆ **11/29: TEST**

◆ **12/06 (Last Lecture)**

Lecture 07: Hydrologic Statistical Frontiers-
Introduction & Examples

◆ **12/13: Final Project Presentation (Details TBA by 12/01)**

Slide 9 Proof

1 Recapitulation

We introduced the method of maximum likelihood for simple linear regression in the notes for two lectures ago. Let's review.

We start with the statistical model, which is the Gaussian-noise simple linear regression model, defined as follows:

1. The distribution of X is arbitrary (and perhaps X is even non-random).
2. If $X = x$, then $Y = \beta_0 + \beta_1 x + \epsilon$, for some constants ("coefficients", "parameters") β_0 and β_1 , and some random noise variable ϵ .
3. $\epsilon \sim N(0, \sigma^2)$, and is independent of X .
4. ϵ is independent across observations.

A consequence of these assumptions is that the response variable Y is independent across observations, conditional on the predictor X , i.e., Y_1 and Y_2 are independent given X_1 and X_2 (Exercise 1).

As you'll recall, this is a special case of the simple linear regression model: the first two assumptions are the same, but we are now assuming much more about the noise variable ϵ : it's not just mean zero with constant variance, but it has a particular distribution (Gaussian), and everything we said was uncorrelated before we now strengthen to independence¹.

Because of these stronger assumptions, the model tells us the conditional pdf of Y for each x , $p(y|X = x; \beta_0, \beta_1, \sigma^2)$. (This notation separates the random variables from the parameters.) Given any data set $(x_1, y_1), (x_2, y_2), \dots, (x_n, y_n)$, we can now write down the probability density, under the model, of seeing that data:

$$\prod_{i=1}^n p(y_i|x_i; \beta_0, \beta_1, \sigma^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(y_i - (\beta_0 + \beta_1 x_i))^2}{2\sigma^2}}$$

¹See the notes for lecture 1 for a reminder, with an explicit example, of how uncorrelated random variables can nonetheless be strongly statistically dependent.

In multiplying together the probabilities like this, we are using the independence of the Y_i .

When we see the data, we do not *known* the true parameters, but any guess at them, say (b_0, b_1, s^2) , gives us a probability density:

$$\prod_{i=1}^n p(y_i|x_i; b_0, b_1, s^2) = \prod_{i=1}^n \frac{1}{\sqrt{2\pi s^2}} e^{-\frac{(y_i - (b_0 + b_1 x_i))^2}{2s^2}}$$

This is the **likelihood**, a function of the parameter values. It's just as informative, and much more convenient, to work with the **log-likelihood**,

$$L(b_0, b_1, s^2) = \log \prod_{i=1}^n p(y_i|x_i; b_0, b_1, s^2) \quad (1)$$

$$= \sum_{i=1}^n \log p(y_i|x_i; b_0, b_1, s^2) \quad (2)$$

$$= -\frac{n}{2} \log 2\pi - n \log s - \frac{1}{2s^2} \sum_{i=1}^n (y_i - (b_0 + b_1 x_i))^2 \quad (3)$$

In the **method of maximum likelihood**, we pick the parameter values which maximize the likelihood, or, equivalently, maximize the log-likelihood. After some calculus (see notes for lecture 5), this gives us the following estimators:

$$\hat{\beta}_1 = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sum_{i=1}^n (x_i - \bar{x})^2} = \frac{c_{XY}}{s_X^2} \quad (4)$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \quad (5)$$

$$\hat{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n (y_i - (\hat{\beta}_0 + \hat{\beta}_1 x_i))^2 \quad (6)$$

As you will recall, the estimators for the slope and the intercept exactly match the least squares estimators. This is a special property of assuming independent Gaussian noise. Similarly, $\hat{\sigma}^2$ is exactly the in-sample mean squared error.

Slide 10 Proof

$$\begin{aligned}\text{Var}(\hat{\beta}_1) &= \text{Var}\left(\frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}\right) \\&= \text{Var}\left(\frac{\sum_i (x_i - \bar{x})(\beta_0 + \beta_1 x_i + u_i)}{\sum_i (x_i - \bar{x})^2}\right), \quad \text{substituting in the above} \\&= \text{Var}\left(\frac{\sum_i (x_i - \bar{x})u_i}{\sum_i (x_i - \bar{x})^2}\right), \quad \text{noting only } u_i \text{ is a random variable} \\&= \frac{\sum_i (x_i - \bar{x})^2 \text{Var}(u_i)}{(\sum_i (x_i - \bar{x})^2)^2}, \quad \text{independence of } u_i \text{ and, } \text{Var}(kX) = k^2 \text{Var}(X) \\&= \frac{\sigma^2}{\sum_i (x_i - \bar{x})^2}\end{aligned}$$